ClipSAT : Multivariable Calculus Partials, Multiple Integrals & Vector Calcu

Unit 5.01: Triple Integrals, Cylindrical & Spherical Coordinates

Name: _____

Q1. Evaluate iiint_B x y z,dV over the box B=[0,3]times[0,1]times[0,3] .

[ ] A: frac{81}{4}
[ ] B: frac{81}{16}
[ ] C: frac{81}{8}
[ ] D: frac{153}{8}

Q2. Evaluate iiint_B x + y + z,dV over the box B=[0,3]times[0,2]times[0,1] .

[ ] A: 36
[ ] B: 9
[ ] C: 18
[ ] D: 24

Q3. Evaluate iiint_B x y z,dV over the box B=[0,2]times[0,1]times[0,2] .

[ ] A: 4
[ ] B: 1
[ ] C: 6
[ ] D: 2

Q4. Evaluate iiint_B y z^{2},dV over the box B=[0,1]times[0,1]times[0,3] .

[ ] A: 9
[ ] B: frac{9}{4}
[ ] C: frac{15}{2}
[ ] D: frac{9}{2}

Q5. Evaluate iiint_B x^{2} z,dV over the box B=[0,1]times[0,1]times[0,3] .

[ ] A: 3
[ ] B: frac{3}{4}
[ ] C: frac{3}{2}
[ ] D: frac{9}{2}

Q6. Evaluate iiint_B x y + z,dV over the box B=[0,1]times[0,1]times[0,3] .

[ ] A: frac{21}{4}
[ ] B: frac{21}{2}
[ ] C: frac{21}{8}
[ ] D: frac{33}{4}

Q7. Evaluate iiint_B x y + z,dV over the box B=[0,1]times[0,3]times[0,3] .

[ ] A: frac{81}{2}
[ ] B: frac{81}{8}
[ ] C: frac{81}{4}
[ ] D: frac{117}{4}

Q8. Evaluate iiint_B x + y + z,dV over the box B=[0,3]times[0,3]times[0,1] .

[ ] A: 63
[ ] B: frac{63}{2}
[ ] C: frac{63}{4}
[ ] D: frac{81}{2}